

LECHFELD, Germany

WMO: 108560

Lat: 48.1850N

Lon: 10.8508E

Elev: 1844

StdP: 13.74

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	13.7	5.4	7.8	8.7	11.2	10.4	15.1	28.7	46.4	25.2	43.6	4.2	180	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	86.3	66.3	82.7	65.1	79.5	64.0	67.8	82.5	66.3	79.8	65.0	77.5	6.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.7	91.2	72.9	61.1	86.3	70.9	60.3	83.6	70.1	33.1	83.0	31.9	80.0	30.9	77.7	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.0	17.8	15.1	DB	1.0	91.5	6.3	3.0	-3.5	93.6	-7.1	95.3	-10.6	97.0	-15.2	99.2	(4)
(5)			WB	0.8	70.9	6.1	2.6	-3.6	72.7	-7.2	74.3	-10.7	75.7	-15.1	77.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.8	30.8	32.8	39.6	47.1	55.2	61.8	64.9	64.3	56.1	48.6	38.6	32.9	(6)
(7)		DBStd	14.11	8.76	9.05	7.66	6.93	6.42	6.56	5.78	5.73	6.09	7.05	7.32	8.03	(7)
(8)		HDD50	2568	597	483	328	134	23	1	0	0	14	112	347	529	(8)
(9)		HDD65	6461	1062	902	787	538	310	138	75	83	271	510	791	994	(9)
(10)		CDD50	1766	0	1	6	46	183	354	460	445	198	68	5	0	(10)
(11)		CDD65	184	0	0	0	0	5	40	70	62	6	1	0	0	(11)
(12)		CDH74	2346	0	0	0	9	130	533	830	744	95	5	0	0	(12)
(13)		CDH80	641	0	0	0	0	17	133	255	228	8	0	0	0	(13)
(14)	Wind	WSAvg	6.6	7.4	7.5	8.0	6.8	6.4	5.8	6.1	5.7	5.8	6.1	6.5	7.1	(14)
(15)	Precipitation	PrecAvg	37.3	2.0	1.9	2.4	2.4	4.1	4.6	4.7	4.3	3.1	2.6	2.6	2.4	(15)
(16)		PrecMax	50.6	4.7	4.0	5.9	5.6	8.4	8.4	9.8	8.7	6.0	6.6	5.9	5.1	(16)
(17)		PrecMin	27.8	0.0	0.6	0.7	0.4	0.3	2.0	0.8	1.6	0.9	0.6	0.0	0.1	(17)
(18)		PrecStd	5.2	1.2	0.9	1.2	1.3	1.9	1.7	1.9	1.9	1.4	1.3	1.4	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	59.1	66.3	75.3	82.5	88.7	90.9	91.7	80.9	73.4	62.3	53.7	(19)
(20)			MCWB	46.3	47.1	50.4	55.8	62.9	67.1	67.9	66.6	63.3	59.5	52.6	46.5	(20)
(21)		2%	DB	50.1	53.5	61.1	70.1	77.5	84.0	86.4	85.8	76.8	68.2	57.1	50.0	(21)
(22)			MCWB	44.3	44.3	48.6	53.8	61.4	66.0	66.4	65.9	62.7	58.0	50.2	45.3	(22)
(23)		5%	DB	46.6	49.5	56.6	66.1	73.5	80.3	82.5	82.0	73.0	64.3	53.1	47.3	(23)
(24)			MCWB	41.8	42.7	46.8	52.7	59.2	64.5	65.4	64.7	61.1	55.8	47.9	43.0	(24)
(25)		10%	DB	43.0	45.7	52.1	62.0	69.5	76.6	78.7	77.9	68.3	60.1	49.7	44.6	(25)
(26)			MCWB	39.6	40.6	44.9	50.4	57.0	62.8	64.1	64.0	59.0	54.1	45.9	41.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	48.2	52.1	57.5	65.1	69.4	69.6	69.2	65.4	60.5	53.5	48.0	(27)
(28)			MCD B	52.2	57.0	63.2	71.7	78.3	84.0	86.1	84.6	76.9	70.2	60.0	51.7	(28)
(29)		2%	WB	44.7	45.5	49.5	55.0	62.4	67.1	67.8	67.3	63.4	58.4	50.9	45.8	(29)
(30)			MCD B	49.1	52.2	58.1	68.0	75.3	80.7	83.1	81.7	74.3	66.7	56.1	49.6	(30)
(31)		5%	WB	42.1	43.0	47.5	53.1	60.1	65.3	66.3	65.8	61.7	56.5	48.4	43.5	(31)
(32)			MCD B	46.0	48.6	55.5	64.1	71.7	78.1	79.7	79.2	71.1	62.8	52.3	46.6	(32)
(33)		10%	WB	39.6	40.5	45.5	51.0	58.0	63.4	64.7	64.4	59.7	54.4	46.2	41.2	(33)
(34)			MCD B	42.7	45.0	51.9	60.5	67.7	74.8	76.8	76.9	67.1	59.5	49.5	44.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	11.9	14.7	17.5	20.8	20.4	20.6	20.6	20.7	18.8	16.8	12.3	11.2	(35)
(36)			MCD BR	16.1	21.8	26.5	30.2	29.2	28.7	29.0	28.9	27.4	24.0	20.0	15.4	(36)
(37)			MCWBR	11.4	14.1	15.9	16.1	14.8	13.3	12.1	12.1	14.1	14.5	13.9	11.1	(37)
(38)		5% WB	MCD BR	14.9	19.5	23.4	27.1	26.5	25.8	25.9	25.7	23.9	21.5	18.3	13.8	(38)
(39)			MCWBR	11.1	13.1	14.5	15.1	13.8	12.3	11.4	11.4	12.8	13.5	13.2	10.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.292	0.316	0.349	0.381	0.391	0.396	0.399	0.393	0.369	0.341	0.312	0.287	(40)	
(41)		taud	2.455	2.389	2.353	2.290	2.298	2.315	2.300	2.335	2.380	2.450	2.486	2.468	(41)	
(42)		Ebn at Noon	251	267	274	275	276	275	272	269	264	254	240	237	(42)	
(43)		Edh at Noon	20	27	33	38	40	39	39	36	32	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	343	621	977	1363	1573	1763	1700	1493	1081	689	372	278	(44)	
(45)		RadStd	45	83	119	180	180	157	155	150	131	81	49	36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.68	N/A	N/A	N/A	N/A	+1.15	N/A	N/A	N/A	N/A
(47)	Regional (84 neighbors)	N/A	N/A	N/A	N/A	+0.72	+0.72	N/A	N/A	+104	+36

Nomenclature: See separate page